FrameCheck: Project Proposal

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Course: BCSE307L Database Systems

Title

FrameCheck: a GPU-accelerated Ethereum transaction-graph analyzer and framing audit

Problem statement

On Ethereum, anyone can send funds to any wallet, and the receiver cannot refuse them. That makes the transaction graph easy to manipulate from outside: an attacker can make an innocent wallet look connected to a hack just by sending it small "dust" transfers. Analysis tools that flag wallets by their proximity to known bad actors can therefore be made to flag someone who did nothing.

FrameCheck audits every flag its detectors raise and answers two questions: **could an outsider have produced this flag without the wallet owner doing anything, and how cheaply?**

A flag is **frameable** if it still holds with every transaction signed by the wallet removed, because unsolicited inbound transfers alone were enough to produce it. It is **robust** if it disappears, because it depended on the wallet's own signed behaviour, which an attacker cannot forge.

Prior work. FrameCheck builds on Liu et al., "Evasion Under Blockchain Sanctions" (WWW '26), which introduced dusting attacks and the proportional impurity score (`impurity_wei / balance_wei`). That score is one of the detectors FrameCheck audits. FrameCheck does not claim to have invented dusting attacks, taint models or counterfactual explanations.

Objectives

1. Ingest a bounded Ethereum subgraph from Etherscan into a normalized PostgreSQL schema.
2. Run PageRank, Louvain and BFS on graph snapshots, on the GPU (cuDF + cuGraph) with a CPU fallback (pandas + NetworkX), and measure the difference honestly.
3. Raise flags with five detectors: BFS proximity, Sybil cluster, PageRank threshold, impurity and engaged proximity.
4. Audit each flag as frameable or robust, with a framing cost where one exists (an upper bound unless the method is `exact`).
5. Make the audit trail tamper-evident in the database itself, using a hash chain, append-only triggers and system-versioned flags.

6. Deliver a secure, role-based web dashboard and desktop app, with CRUD, search and pagination, deployed in the cloud.

Scope

In scope

- Ethereum mainnet, native ETH and ERC-20 transfers, through a bounded 2-hop crawl around a case-study address.
- A PostgreSQL schema of 25 tables, with triggers, functions, a view, a materialized view and stored procedures.
- SQLAlchemy ORM data access, with DDL in SQL migrations.
- Google sign-in and bcrypt local accounts, with admin, analyst and viewer roles.
- A FastAPI backend, a browser dashboard and a Tauri desktop app.
- Backup and verified restore with `pg_dump` / `pg_restore`.
- Docker, docker compose and cloud deployment.
- A framing **simulator** that only inserts synthetic rows into a local database, tagged `is_simulated`.

Out of scope

- Internal (contract-initiated) transactions. Ingestion never calls Etherscan's `txlistinternal`, so these transfers are invisible to the pipeline and to every audit built on it. This is a known limitation.
- Other blockchains.
- Crawls beyond the configured hop and transaction limits.
- Sending real transactions. Framing is simulation only.
- Attributing wallets to real-world identities.
- Exact minimum framing costs in general: greedy results are reported as upper bounds.

Team members and responsibilities

Member	Register number	Responsibilities
Vishal Muthukrishnan	24BCE2802	`<responsibilities>`
Shalom Santosh	24BCE2731	`<responsibilities>`

Technology stack (as built)

Layer	Technology
Database	PostgreSQL 16+, PL/pgSQL triggers, functions and procedures
ORM	SQLAlchemy 2.x over psycopg2

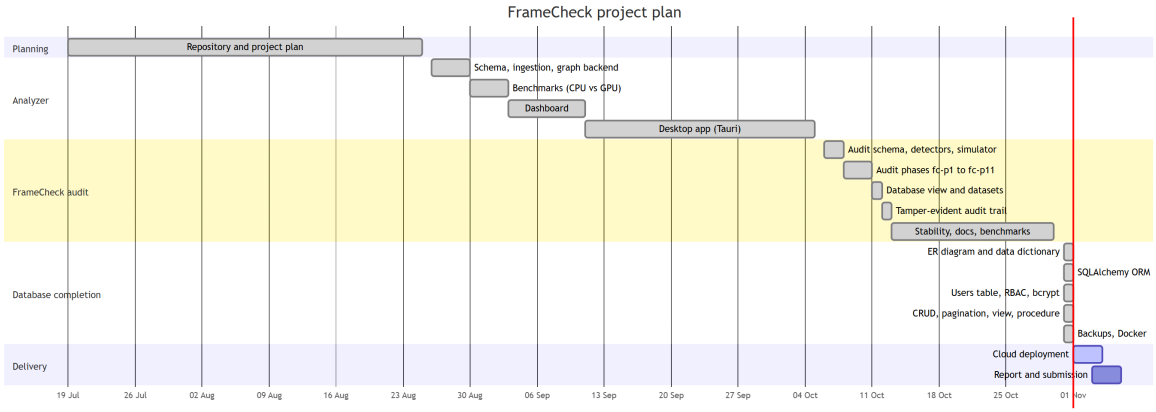
Layer	Technology
Backend	Python, FastAPI, Uvicorn
Graph analytics	cuDF + cuGraph (GPU, RAPIDS in WSL2); pandas + NetworkX (CPU)
Frontend	HTML, CSS and JavaScript with no build step; WebGL graph view
Desktop	Tauri 2 (Rust) with a PyInstaller-frozen backend sidecar; wgpu layout
Authentication	Google OAuth 2.0 (PKCE), bcrypt local accounts, role-based access control
DevOps	Docker, docker compose, Docker Hub (private), GitHub Actions, Render, Neon

Major deliverables

Deliverable	Where
Source code	GitHub repository `LIVIDFOX1116/DBMS` (private), branch `dev`
Schema DDL and migrations	`db/schema.sql`, `db/migrations/`
ER diagram	`ER_Diagram_TransactionGraphAnalyzer.svg` / `.png`, from `schema_v2.dbml`
Data dictionary	`DATA_DICTIONARY.md`
ORM layer	`db/orm/`
API reference, ORM coverage	`API_CONTRACT.md`
Audit method and results	`FRAMECHECK.md`, `RESULTS.md`
GPU vs CPU benchmarks	`benchmark_results.md`
Backup and restore	`BACKUP_RESTORE.md`, `scripts/backup_db.*`, `scripts/restore_db.*`
Container image	`Dockerfile`, `docker-compose.yml`
Desktop app	`src-tauri/` (Windows installer)
Live URL	https://framecheck-awlf.onrender.com (sign-in required)

Gantt chart

Completed work uses actual dates from git history and tags (`fc-p1` to `fc-p11`). Deployment and submission are planned dates.


FrameCheck project plan
Planning
Repository and project plan
Schema, ingestion, graph backend
Analyzer
Benchmarks (CPU vs GPU)
Dashboard
Desktop app (Tauri)
Audit schema, detectors, simulator
Audit phases fc-p1 to fc-p11
FrameCheck audit
Database view and datasets
Tamper-evident audit trail
Stability, docs, benchmarks
ER diagram and data dictionary
SQLAlchemy ORM
Database completion
Users table, RBAC, bcrypt
CRUD, pagination, view, procedure
Backups, Docker
Delivery
Cloud deployment
Report and submission
19 Jul
26 Jul
02 Aug
09 Aug
16 Aug
23 Aug
30 Aug
06 Sep
13 Sep
20 Sep
27 Sep
04 Oct
11 Oct
18 Oct
25 Oct
01 Nov

Gantt chart